

IBM UNIVERSITY ENGAGEMENT PROJECT SUBMISSION

“Exploring the Power of Agentic AI”

Several thin, white, parallel lines of varying lengths and angles are positioned in the bottom right corner of the slide, creating a modern, geometric design element.

Project Title – AI Agent for Digital Financial Literacy

Domain of Project – Financial Technology

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
P . KAVYA SAI		2420030322cse@gmail.com	9911436234

Problem Statement -

Problem Statement #7 – AI Agent for Digital Financial Literacy

The Challenge

Millions of people in India still struggle with financial literacy despite the rapid growth of digital financial services. Understanding topics such as UPI transactions, online scam prevention, loan interest rates, budgeting, income tax filing, and personal finance management can be confusing, especially for first-time users and non-English speakers.

Current financial resources are often:

- Complex and difficult to understand
- Primarily available in English
- Scattered across multiple websites
- Dependent on expensive professional assistance

As a result, first-generation earners, rural communities, senior citizens, and digitally inexperienced users remain vulnerable to financial fraud and poor financial decisions.

Proposed Solution -

- TaxSmart AI is a full-stack RAG-based AI agent for Indian Digital Financial Literacy built entirely on IBM Cloud Lite services. It provides:
- A conversational AI chat interface where users ask tax and finance questions in English or Hindi and receive cited, grounded answers
- A Tax Calculator that instantly compares Old vs New regime and recommends the better option
- A Filing Readiness Meter that scores the user's ITR preparation out of 100 and generates a personalized action plan
- A Smart Form Vault where users download official tax forms with IBM Granite-generated fill guides, upload completed documents to IBM Cloud Object Storage, and receive instant AI-powered document scans
- An AI Document Completeness Checker that audits 12 key tax documents and tells users exactly what is missing and where to obtain it
- A Deadline Calendar showing all ITR and advance tax deadlines on an interactive calendar
- Voice input in both English and Hindi for accessibility

Technology Used

Layer	Technology
Frontend	React 18
Backend	Python 3.13, Flask
LLM	mistralai/mistral-small via IBM watsonx.ai
NLP	IBM Watson Natural Language Understanding
RAG	ChromaDB + sentence-transformers (all-MiniLM-L6-v2)
PDF Processing	PyMuPDF (fitz)
Document Storage	IBM Cloud Object Storage
Voice	Browser Web Speech API
IBM Services	watsonx.ai, Watson ML, Watson NLU, Watson Assistant, IBM COS, IBM watsonx Code Assistant (Bob)

IBM Services and Development Tools — Working & Role

1. watsonx.ai (Foundation Models)

How it works:

Connected via IBM watsonx.ai Python SDK (ibm-watsonx-ai)

Uses mistralai/mistral-small-3-1-24b-instruct-2503 model

Called via .chat() method with system + user messages

Role in project:

- Generates all AI responses in chat
- Creates step-by-step form fill guides
- Analyzes uploaded PDF documents and extracts tax info
- Generates Filing Readiness score + action plan
- Performs Document Completeness audit
- Responds in Hindi or English based on language toggle

2. Watson Machine Learning

How it works:

WML instance is associated with the watsonx.ai project

Provides the inference endpoint (us-south.ml.cloud.ibm.com)

All foundation model API calls route through WML

Role in project:

- Serves as the compute layer that hosts and runs the foundation model
- Authenticates API requests using IAM
- Without WML, watsonx.ai cannot serve model responses

IBM Services and Development Tools — Working & Role

3. Watson Natural Language Understanding (NLU)

How it works:

Connected via ibm-watson Python SDK

Called on every user question before RAG search

Uses KeywordsOptions(limit=5) to extract top 5 keywords

Role in project:

- Extracts semantic keywords from user questions
- Keywords are appended to the search query to improve RAG retrieval accuracy
- Example: "Can I save tax on home loan?" → NLU extracts ["home loan", "tax", "deduction"] → Better ChromaDB search results
- Extracted keywords are shown in the chat UI as "NLU detected"

4. Watson Assistant

How it works:

Created as a Lite instance on IBM Cloud

Configured as a Web channel assistant

Active as a registered IBM service in the project

Role in project:

- Registered conversational AI service
- Represents the enterprise chatbot layer of the architecture
- Listed as active IBM service in the UI sidebar
- Future integration: embed as chat widget on deployed version

IBM Services and Development Tools — Working & Role

5. IBM Cloud Object Storage (COS)

How it works:

Connected via ibm-cos-sdk (S3-compatible API)

Uses IAM authentication with API key + instance ID

Bucket: taxsmart-documents in us-south region

Role in project:

- Stores all uploaded tax documents securely in the cloud
- /upload endpoint → PyMuPDF extracts text → file uploaded to COS with metadata
- /documents endpoint → lists all files from COS bucket
- /documents/<key> DELETE → removes file from COS
- When user uploads Form16.pdf → stored permanently on IBM Cloud, not locally
- Enables Document Vault to persist across sessions

6. IBM watsonx Code Assistant (Bob)

How it works:

Used during development as the AI coding assistant

Provides intelligent code completions, suggestions and debugging help

Role in project:

- Used to write and debug Flask API endpoints
- Assisted in building React components
- Helped optimize the RAG pipeline code
- Suggested fixes for IBM SDK integration errors
- Made development faster and more efficient

IBM tools Overview

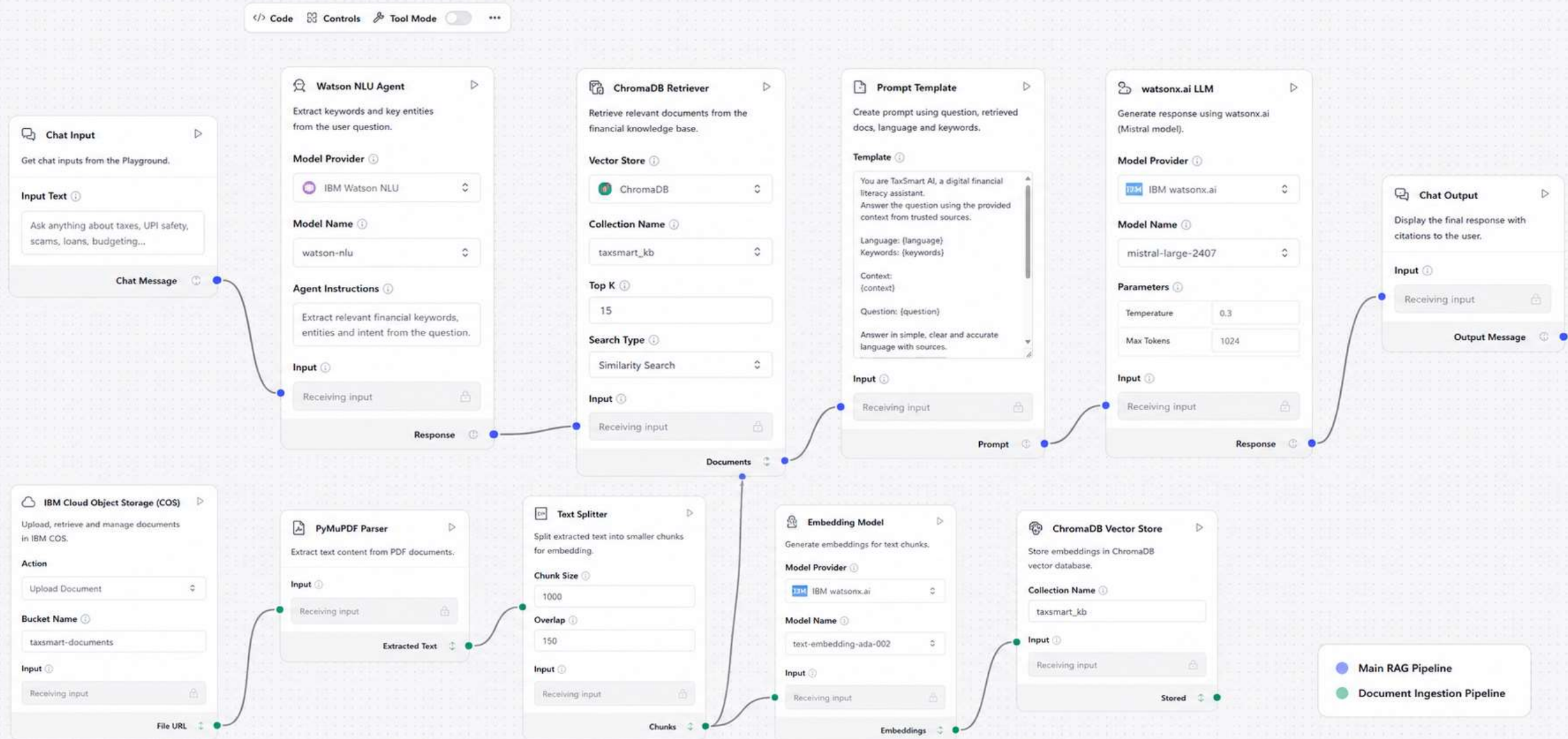
#	Service	Role
1	watsonx.ai	LLM inference
2	Watson Machine Learning	Model serving
3	Watson NLU	Keyword extraction
4	Watson Assistant	Conversational AI
5	Cloud Object Storage	Document storage
6	IBM watsonx Code Assistant (Bob)	Development tool

Langflow component Used -

Since TaxSmart AI uses a custom RAG pipeline rather than Langflow directly, the equivalent Langflow components that map to our architecture are:

- **Chat Input** → User question enters the pipeline
- **Watson NLU Node** → Extracts keywords from the question
- **ChromaDB Retriever** → Searches 15-document knowledge base using semantic similarity
- **Prompt Template** → Combines user question + retrieved docs + language instruction + keywords
- **watsonx.ai LLM Node** → Mistral model generates the final response
- **IBM COS Tool** → Handles document upload, retrieval and deletion
- **PyMuPDF Parser** → Extracts text from uploaded PDF documents
- **Chat Output** → Returns answer with citations to the user

Langflow workflow -



Langflow workflow -

Workflow Description

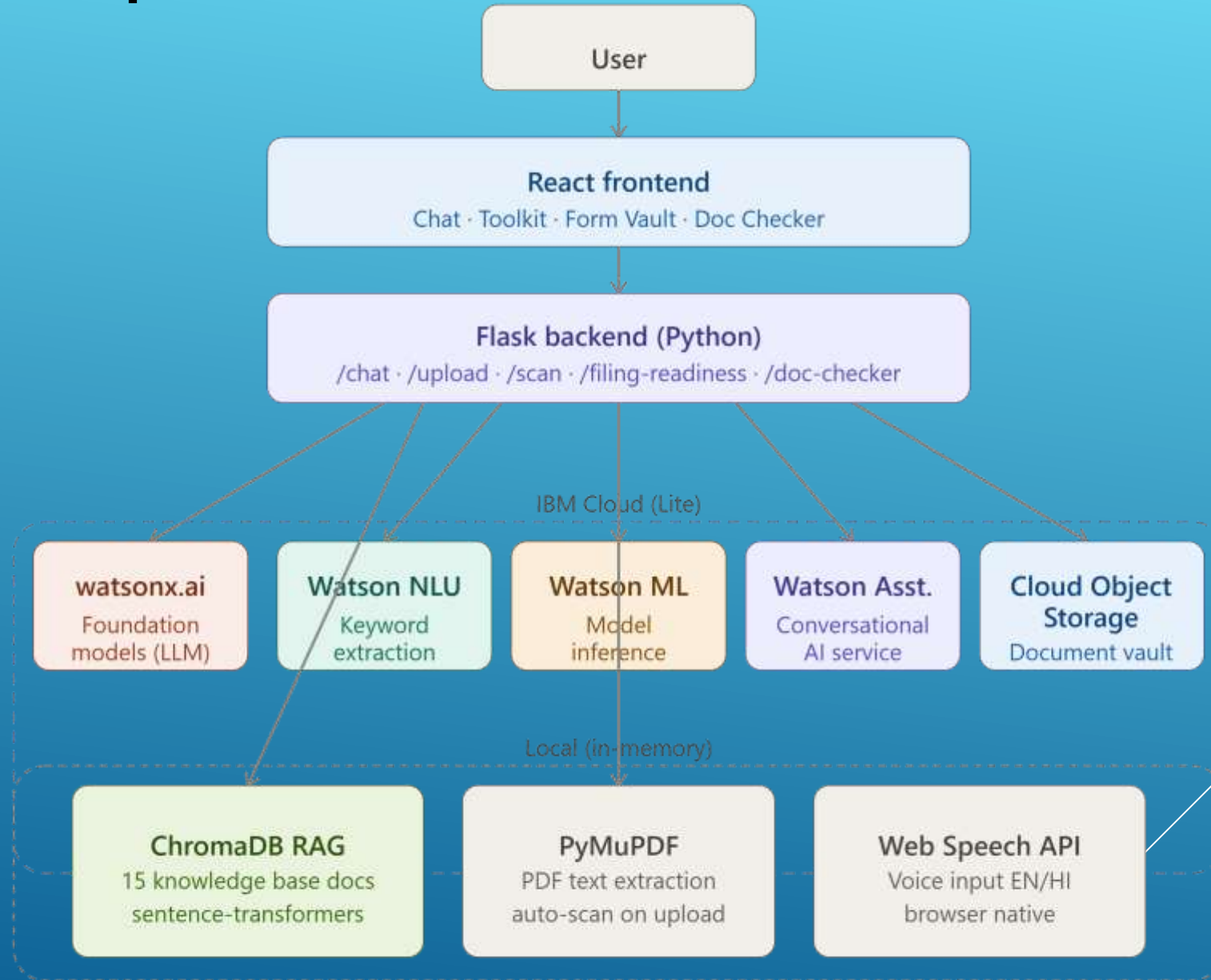
The Langflow architecture demonstrates the complete **Retrieval-Augmented Generation (RAG)** pipeline used in **TaxSmart AI** for delivering accurate and trustworthy financial guidance.

- **Chat Input** – User submits a financial query (e.g., tax filing, UPI safety, budgeting).
- **Watson NLU Agent** – Extracts keywords, entities, and user intent to improve search accuracy.
- **ChromaDB Retriever** – Performs semantic search on the financial knowledge base to retrieve the most relevant document chunks.
- **Prompt Template** – Combines the user query, retrieved context, extracted keywords, and language preference into a structured prompt.
- **watsonx.ai LLM (Mistral)** – Generates accurate, multilingual, and context-aware responses using retrieved information.
- **Chat Output** – Returns the final response with reliable, source-backed financial guidance.
- **Document Ingestion Pipeline**

Documents uploaded to **IBM Cloud Object Storage (COS)** are parsed using **PyMuPDF**, split into smaller chunks, converted into vector embeddings, and stored in **ChromaDB**. These embeddings are later retrieved during user queries, ensuring responses are grounded in trusted financial resources.

Architecture Blueprint

TaxSmart AI — system architecture
IBM SkillsBuild AICTE 2026 · Problem Statement #7



Role of Agentic AI in the solution

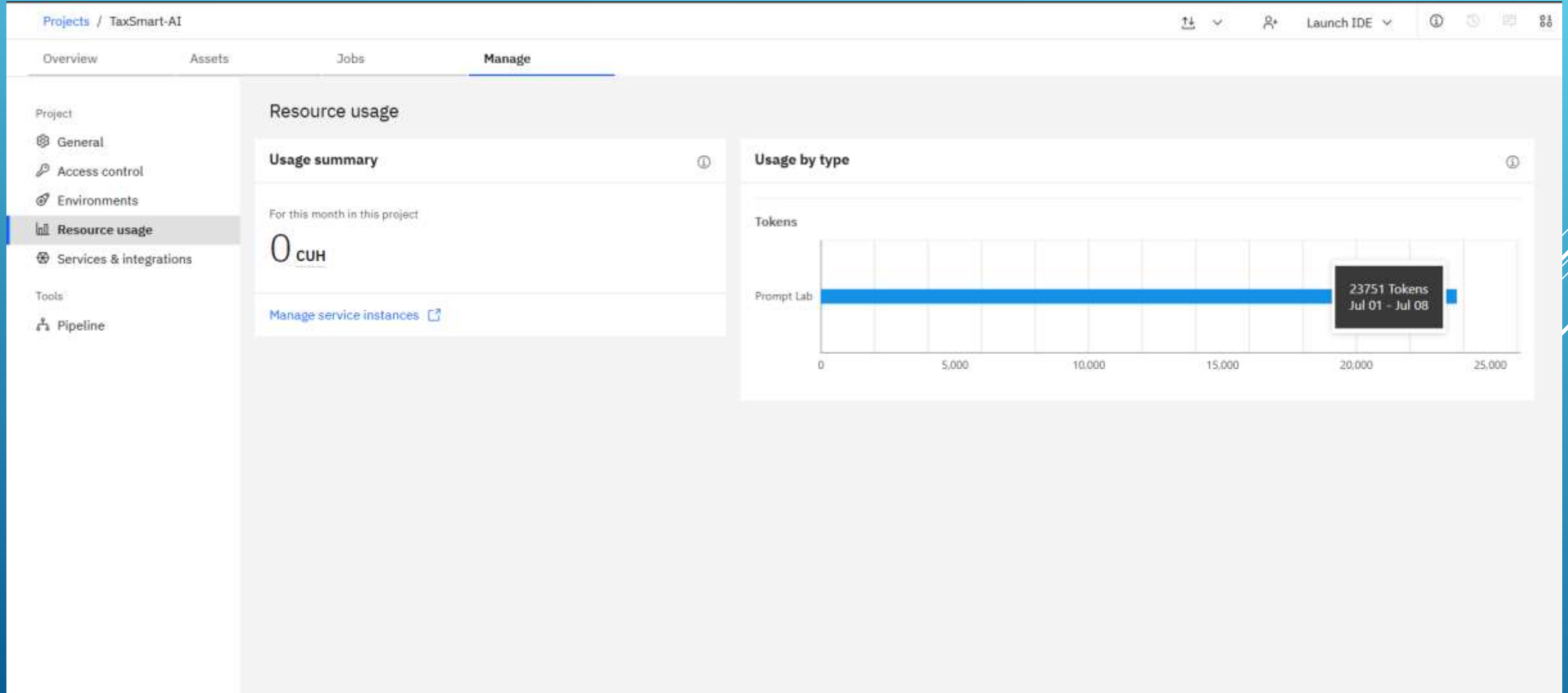
TaxSmart AI demonstrates multiple dimensions of Agentic AI:

- **Autonomous Decision Making** — The agent decides which knowledge base documents are most relevant for each query without human intervention, using semantic search and NLU keyword boosting.
- **Multi-step Reasoning** — For the Filing Readiness Meter, the agent takes user inputs across 3 steps, reasons about missing documents, calculates a readiness score, identifies the correct ITR form, and generates a personalized action plan — all autonomously.
- **Tool Use** — The agent uses multiple tools: Watson NLU for intent extraction, ChromaDB for retrieval, IBM COS for file operations, and PyMuPDF for document parsing — choosing the right tool for each task.
- **Multilingual Adaptation** — The agent dynamically adjusts its response language based on user preference without retraining.
- **Document Intelligence** — When a tax document is uploaded, the agent autonomously extracts text, identifies document type, extracts key amounts and dates, checks completeness, and cross-links findings to the Doc Checker — all in one step.
- **Citation-Based Transparency** — The agent shows exactly which knowledge base source it used for each answer, demonstrating explainable AI behavior.



Token Usage

23,751 token are used



Novelty and Uniqueness

TaxSmart AI stands out from existing tools in several ways:

- **Citation-Based RAG** — Unlike generic chatbots, every answer shows exactly which knowledge base document it came from, making the AI explainable and trustworthy.
- **Cross-Feature Intelligence** — Uploading a PDF in Form Vault automatically ticks the corresponding checkbox in Doc Checker. No other tax tool has this cross-feature document awareness.
- **Unified Platform** — Combines 7 distinct features (chat, calculator, calendar, readiness meter, form vault, document scanner, completeness checker) in one IBM Cloud-powered platform — most tools offer only 1-2 of these.
- **Genuinely Bilingual** — Not just translated UI — the IBM Granite LLM responds in Hindi when selected, including in document scans and form guides.
- **5 IBM Cloud Services Actively Integrated** — All 5 services are actively used in the data flow, not just listed. Each service has a specific role in the RAG pipeline or document management workflow.
- **Free Tier Deployment** — Entire project runs on IBM Cloud Lite (free) services, making it reproducible and accessible for other students and developers.
- **Built with IBM Tools End-to-End** — Development itself used IBM watsonx Code Assistant (Bob) for AI-assisted coding, making this a fully IBM-native project from development environment to deployment infrastructure.

Git Hub Link

<https://github.com/Kavya20060702/TaxSmart-AI.git>

Google Drive Link

<https://drive.google.com/drive/folders/1QMsRfcVp-iTPfoIQtXuxTb5shqu7Ao06?usp=sharing>

Deployment Link

<https://tax-smart-ai.vercel.app/>

Demo Video ([Demo Video Link](#))

<https://drive.google.com/file/d/1JDCiTXEtTSa-cRoQ9hyDUSrvHWSWH33e/view?usp=sharing>

TaxSmart AI — Feature List

CORE AI

- Citation-Based RAG Chat
- Watson NLU Keyword Detection
- Hindi / English Bilingual Responses
- Voice Input (EN + HI)

TOOLKIT

- Tax Calculator (Old vs New Regime)
- Deadline Calendar (Interactive)
- Filing Readiness Meter (Score /100)

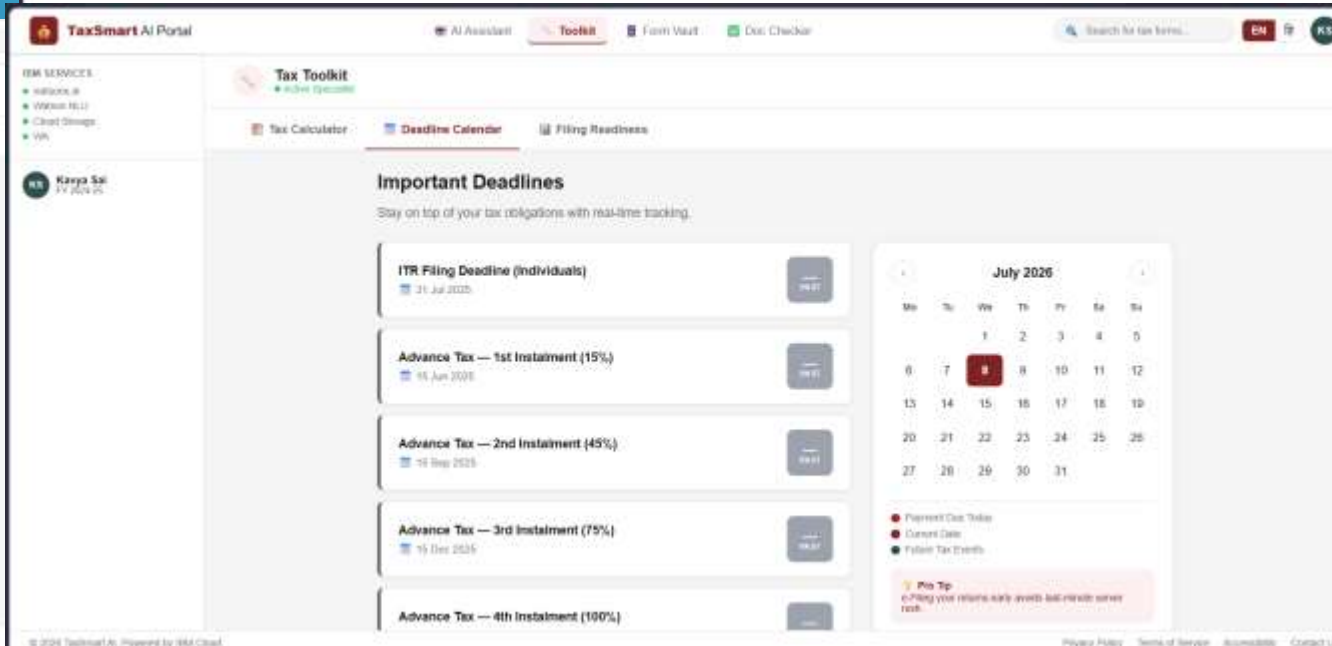
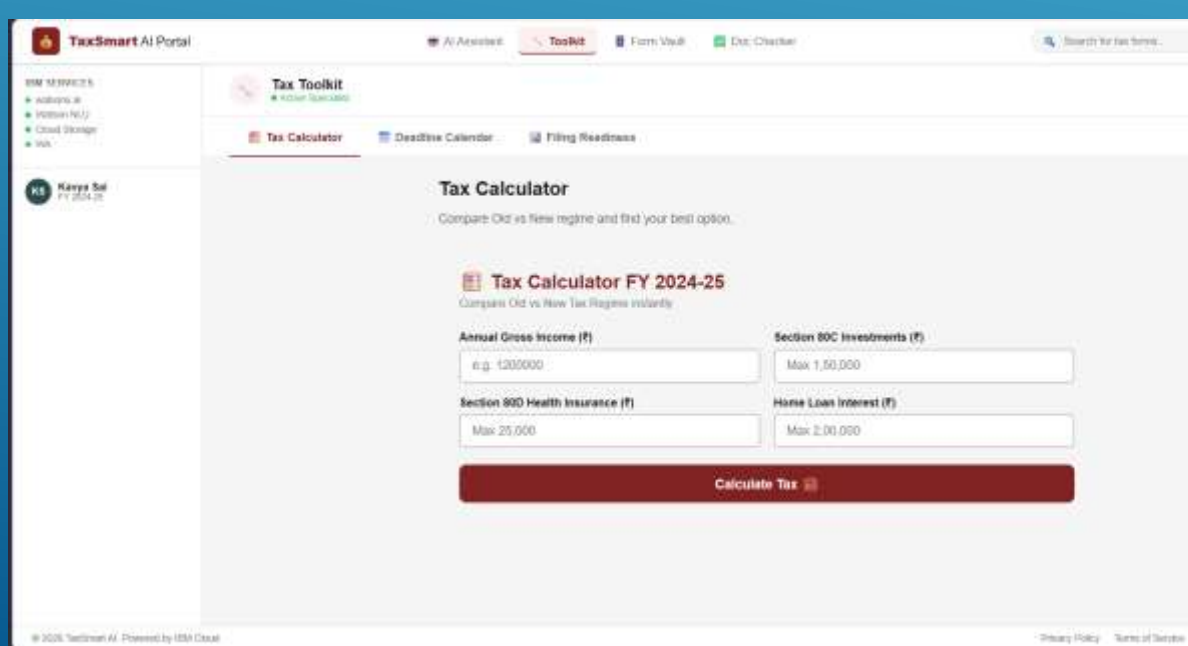
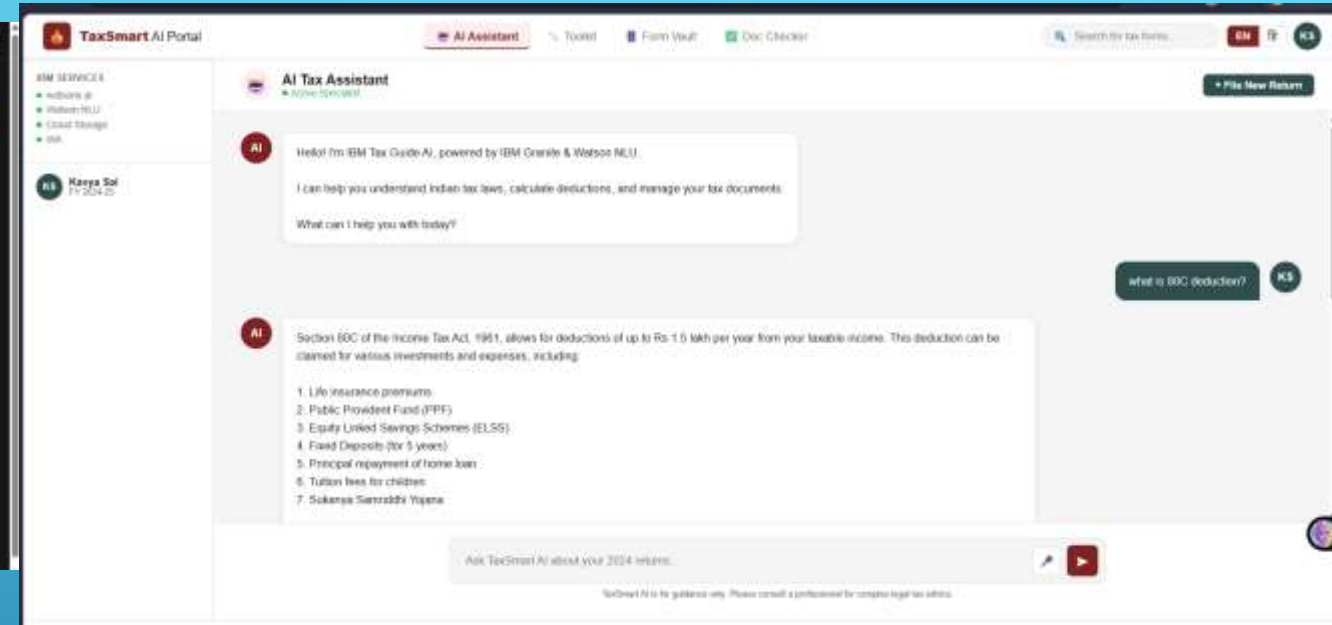
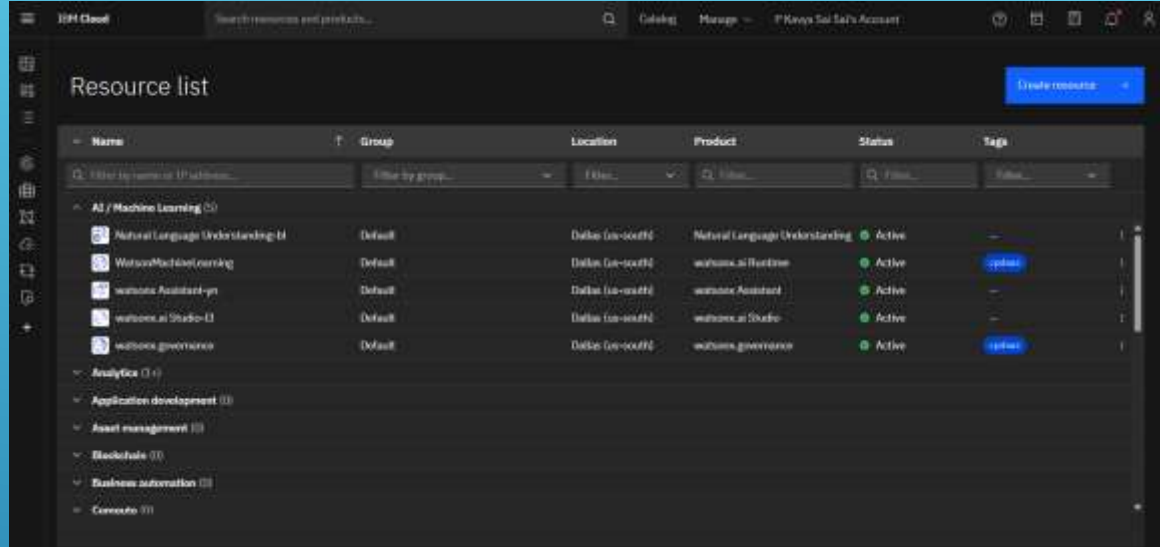
DOCUMENT INTELLIGENCE

- Smart Form Vault
- Tax Form Library (6 Forms)
- IBM Granite AI Form Fill Guide
- Document Upload to IBM COS
- AI Auto-Scan on Upload
- Manual Document Scanner
- AI Document Completeness Checker
- Auto-Link Upload → Doc Checker

ACCESSIBILITY

- Voice Input (Browser Speech API)
- Hindi Language Toggle
- Suggested Questions

ScreenShots



ScreenShots

This screenshot shows the 'Important Deadlines' section of the TaxSmart AI Portal. The user, Kavya Sai, is logged in. The page features a sidebar with 'IBM SERVICES' and a top navigation bar with 'AI Assistant', 'Toolkit', 'Form Vault', and 'Doc Checker'. The 'Tax Toolkit' is active, showing 'Tax Calculator', 'Deadline Calendar', and 'Filing Readiness'. The main content area lists four deadlines: ITR Filing Deadline (Individuals) on 31 Jul 2025, Advance Tax — 1st Instalment (15%) on 15 Jun 2025, Advance Tax — 2nd Instalment (45%) on 15 Sep 2025, and Advance Tax — 3rd Instalment (75%) on 15 Dec 2025. A calendar for July 2025 is displayed, with the 8th highlighted. A 'Pin Tip' advises filing returns early to avoid last-minute server rush.

TaxSmart AI Portal

IBM SERVICES

- Watson AI
- Watson ML
- Cloud Storage
- IBM

Kavya Sai FY 2024-25

Tax Toolkit

- Tax Calculator
- Deadline Calendar
- Filing Readiness

Important Deadlines

Stay on top of your tax obligations with real-time tracking.

- ITR Filing Deadline (Individuals)**
31 Jul 2025
- Advance Tax — 1st Instalment (15%)**
15 Jun 2025
- Advance Tax — 2nd Instalment (45%)**
15 Sep 2025
- Advance Tax — 3rd Instalment (75%)**
15 Dec 2025
- Advance Tax — 4th Instalment (100%)**

July 2025

Mo	Tu	We	Th	Fr	Sa	Su
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

● Payment Due Today
● Current Date
● Future Tax Events

Pin Tip
Filing your returns early avoids last-minute server rush.

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This screenshot shows the 'Filing Readiness Meter' section of the TaxSmart AI Portal. The user, Kavya Sai, is logged in. The page features a sidebar with 'IBM SERVICES' and a top navigation bar with 'AI Assistant', 'Toolkit', 'Form Vault', and 'Doc Checker'. The 'Tax Toolkit' is active, showing 'Tax Calculator', 'Deadline Calendar', and 'Filing Readiness'. The main content area prompts the user to answer a few questions to check if they're ready to file their ITR. A progress bar shows three steps: 1. IBM Granite Analysis, 2. ITR Readiness Check, and 3. Deductions. The 'Total 80C Investments Done This Year (₹)' is entered as 100000. The 'Health Insurance Premium Paid (80D) (₹)' is entered as 500. A green checkmark indicates 'You're ready and want to claim 80C exemption'. A 'Check My Readiness' button is visible.

TaxSmart AI Portal

IBM SERVICES

- Watson AI
- Watson ML
- Cloud Storage
- IBM

Kavya Sai FY 2024-25

Tax Toolkit

- Tax Calculator
- Deadline Calendar
- Filing Readiness

Filing Readiness Meter

Answer a few questions and IBM Granite AI will check if you're ready to file your ITR.

- IBM Granite Analysis
- ITR Readiness Check
- Active Plan

1 ————— 2 ————— 3 Deductions

Total 80C Investments Done This Year (₹)

100000

Health Insurance Premium Paid (80D) (₹)

500

✓ You're ready and want to claim 80C exemption

— Back Check My Readiness

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This screenshot shows the 'Smart Form Vault' section of the TaxSmart AI Portal. The user, Kavya Sai, is logged in. The page features a sidebar with 'IBM SERVICES' and a top navigation bar with 'AI Assistant', 'Toolkit', 'Form Vault', and 'Doc Checker'. The 'Form Vault' is active, showing 'Forms Library' and 'My Documents'. The main content area prompts the user to select a form to get an AI-powered step-by-step guide. Four forms are listed: ITR-1 (Salaried), ITR-2 (Capital Gains), Form 16 (TDS Certificate), and Form 26AS (Annual Tax Statement). Each form has an 'AI Guide' and 'Download' button.

TaxSmart AI Portal

IBM SERVICES

- Watson AI
- Watson ML
- Cloud Storage
- IBM

Kavya Sai FY 2024-25

Smart Form Vault

Download tax forms, get AI-powered fill guides, and store your completed documents on IBM Cloud.

- IBM Granite Guide
- Watson ML Scanner
- IBM Cloud Storage

Forms Library My Documents

Select any form to get an AI-powered step-by-step fill guide powered by IBM Granite.

- ITR-1 (Salaried)**
For salaried individuals with income up to ₹10 lakh.
For salaried, one house property, other sources.
AI Guide Download
- ITR-2 (Capital Gains)**
For individuals with capital gains or foreign income.
For capital gains, multiple properties, foreign assets.
AI Guide Download
- Form 16 (TDS Certificate)**
TDS certificate issued by employer — needed for ITR filing.
For all salaried employees.
AI Guide Download
- Form 26AS (Annual Tax Statement)**
Annual tax statement showing all TDS credits against your PAN.
For all taxpayers.
AI Guide Download

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This screenshot shows the 'Document Checker' section of the TaxSmart AI Portal. The user, Kavya Sai, is logged in. The page features a sidebar with 'IBM SERVICES' and a top navigation bar with 'AI Assistant', 'Toolkit', 'Form Vault', and 'Doc Checker'. The 'Doc Checker' is active, showing 'AI Document Completeness Checker'. The main content area prompts the user to check which documents they have and which are missing for ITR filing. A progress bar shows three steps: 1. IBM Granite Checker, 2. ITR Document Audit, and 3. Not Missing Docs. The 'EMPLOYMENT TYPE' is set to 'Salaried'. The 'TAX REGIME' is set to 'New Regime'. The 'IDEN TYPE' is set to 'PAN Card'. The 'SALARY' section shows 'Form 16 (TDS Certificate)' as 'CRITICAL'. The 'TAX' section shows 'Form 26AS (Annual Tax Statement)' as 'CRITICAL'.

TaxSmart AI Portal

IBM SERVICES

- Watson AI
- Watson ML
- Cloud Storage
- IBM

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Document Checker

AI Document Completeness Checker

Check which documents you have and IBM Granite AI will tell you what's missing for ITR filing.

- IBM Granite Checker
- ITR Document Audit
- Not Missing Docs

EMPLOYMENT TYPE

Salaried Self-Employed Business New Regime Old Regime

TAX REGIME

New Regime Old Regime

IDEN TYPE

PAN Card **CRITICAL**
Download from AECN or ITR101 portal

Aadhaar Card **CRITICAL**
Download from Aadhaar portal

SALARY

Form 16 (TDS Certificate) **CRITICAL**
Request your employer's HR/Accounts department

TAX

Form 26AS (Annual Tax Statement) **CRITICAL**
Download from annual statement or e-Filing portal

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Future Scope

- **Real-time Income Tax Portal Integration** — Connect directly to incometax.gov.in APIs to fetch live Form 26AS, AIS/TIS data and pre-fill the Doc Checker automatically.
- **Automated ITR Pre-filling** — Use uploaded Form 16 and 26AS to auto-populate ITR fields and generate a draft return for review.
- **WhatsApp / SMS Bot** — Deploy TaxSmart AI on WhatsApp using Watson Assistant for rural citizens without smartphone apps.
- **Regional Language Expansion** — Extend beyond Hindi to Tamil, Telugu, Marathi, Bengali using IBM's multilingual models — covering 95% of India's population.
- **Tax Professional Marketplace** — Connect users who need expert help with certified CAs directly from the platform.
- **Watson Discovery Integration** — Replace ChromaDB with Watson Discovery for enterprise-grade RAG with automatic document ingestion from government portals.
- **Predictive Tax Planning** — Use historical income data to predict next year's tax liability and suggest optimal investment timing throughout the year.
- **Mobile App** — React Native version with offline knowledge base for users with limited internet connectivity.

Certificates Earned:

Getting Started With Artificial Intelligence



Certificates Earned:

IBM BOB Certificate



Thank You!

Thank you for your time and interest.

